Pratyush Acharya

B.Tech Cyber Physical Systems, MIT Manipal • Embedded systems, drones and CPS security
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Education

Manipal Institute of Technology, MAHE, Manipal Jul 2025 – Jul 2029 (expected)
B.Tech, Cyber Physical Systems, School of Electrical Engineering CGPA: 7.79/10

Research (manuscripts)

Stealthy False-Data Injection on UAV Battery Sensing and a Powertrain-Consistency Detector for Medical Delivery Drones
P. Acharya, Srinivas G. (School of Mechanical Engineering, MIT)

- Characterised when model-consistent "virtual battery" attacks are undetectable from battery telemetry alone; proposed a powertrain-consistency detector using ESC duty cycles and rotor speeds.
- In 2,500 simulated delivery missions per condition, caught every phantom and hidden-drain attack (median 3.0 s and 5.5 s) at a 0.34% false-alarm rate.

Bilinear Actuation Watermarking: Authenticating Low-Cost PM2.5 Sensors in Smart Air Purifiers
P. Acharya (sole author)

- Secret randomised fan-speed challenge with a gain-invariant log-slope test, CUSUM and GLRT timing checks.
- In 1,000 simulated rooms per condition, detected 90–93% of replay attacks within 6 h and all stuck-at attacks, at about 1.0 W and 1.5 dB(A) cost.

Experience

Project DronAid – Electronics & Control Systems Member Feb 2026 – present (9 months)

- Student team building autonomous drones for medical aid delivery and disaster surveillance and recovery; preparing for NIDAR 2027 (RescueSwarm problem statement).
- Working with the team towards a 30 km+ autonomous medical delivery attempt.
- Designed the drone's power distribution and control board: an ESP32 runs the lighting system and relays act as a failsafe kill switch for the motors.
- PCB design in KiCAD and Proteus; led sponsor outreach to 13+ companies.

Class Representative, School of Electrical Engineering Aug 2026 – present
Block Representative, hostel Sep 2026 – present

Projects

Rubik's Cube Solver – C++, WebAssembly • live demo

- Kociemba two-phase solver from scratch with no dependencies; 20 moves or fewer, median 54 ms; runs in-browser via WASM with a 3D cube.

Which Way? – Java, Android

- Fully offline compass to nearby places using GPS, rotation-vector sensor and bundled OpenStreetMap data; Bluetooth LE mesh shares pins between friends; ∼50 KB APK.

PuriSwitch: Smart Air Purifier Switch – ESP32, team of 3

- Smoke, DHT11 and MQ7 sensing with an ESP32 soft-AP web dashboard.

Contactless Water Level Sensor – ESP32, C++

- Capacitive sensing through non-conductive container walls, with calibration and a live Wi-Fi web view.

Skills

Programming: C, C++, Java, Python **Embedded:** ESP32, microcontrollers, sensors, PCB design (KiCAD, Proteus)
Robotics & control: ROS, ArduPilot, MATLAB, Simulink **Other:** Linux, Git, Tinkercad

Certifications & Activities

ROS Workshop, IE Mechatronics Students' Chapter (Apr 2026) • MATLAB Onramp, MathWorks
Team RoboManipal (task phase) • Competitive programming (Codeforces: ginini) • Photography and videography